TABLETS

They're solid dosage form containir drug with or without excipients and prepared by compression. It's the most popular dosage form; it takes 70% of all dosage forms production

The advantages of tablets:

1. Accuracy of the dosage.
2. Ease of administration.
3. More stable than liquid dosage forms
4. Easy to transport in bulk& easily carrie
5. the patient.
   The release rate of the drug can be contro
6. Tablets can be mass-produced simply & quickly, therefore manufacturing cost is lower than other dosage forms.

Shape, size & weight of the tablets:

Size of the tablet:

diameter of tablets = 6mm → 14 mm.

Weight of the tablet:

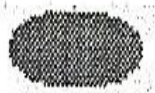
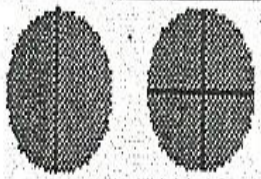
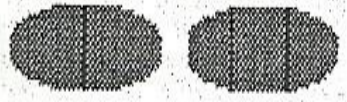
6 mm ⟶ 60mg ~ 120mg.

14 mm ⟶ 800mg ~ 1000mg.

N.B.:

-Thickness must be compatible with the diameter.

-If the weight of the drug is higher than 600mg, manufacturers prefer to use capsules instead of tablets because it's easier to swallow.

Shape	 Oblong	 Round	 Oval	 Square	 Triangle
Surface	 Flat	 Concave	 Deep Concave	 scored tablets	 Scored tablets

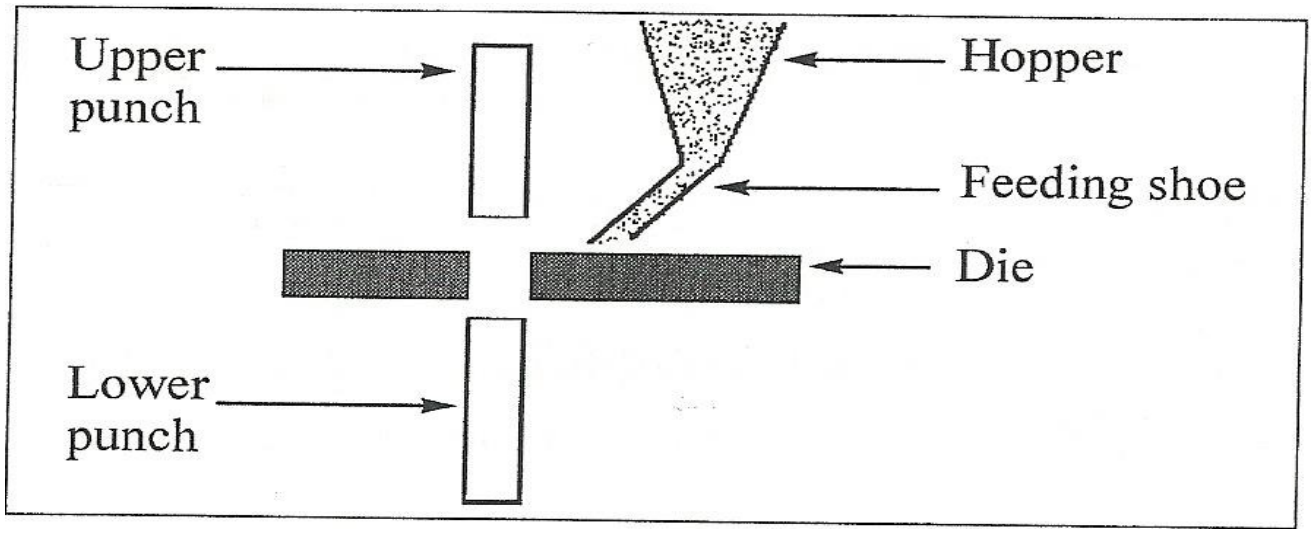
Tablet Press or machine:

It's the basic mechanical unit of all tablets. The press includes:

a) Hopper: for storing the powder to be compressed

b) Die: for controlling the shape and size of the tablet

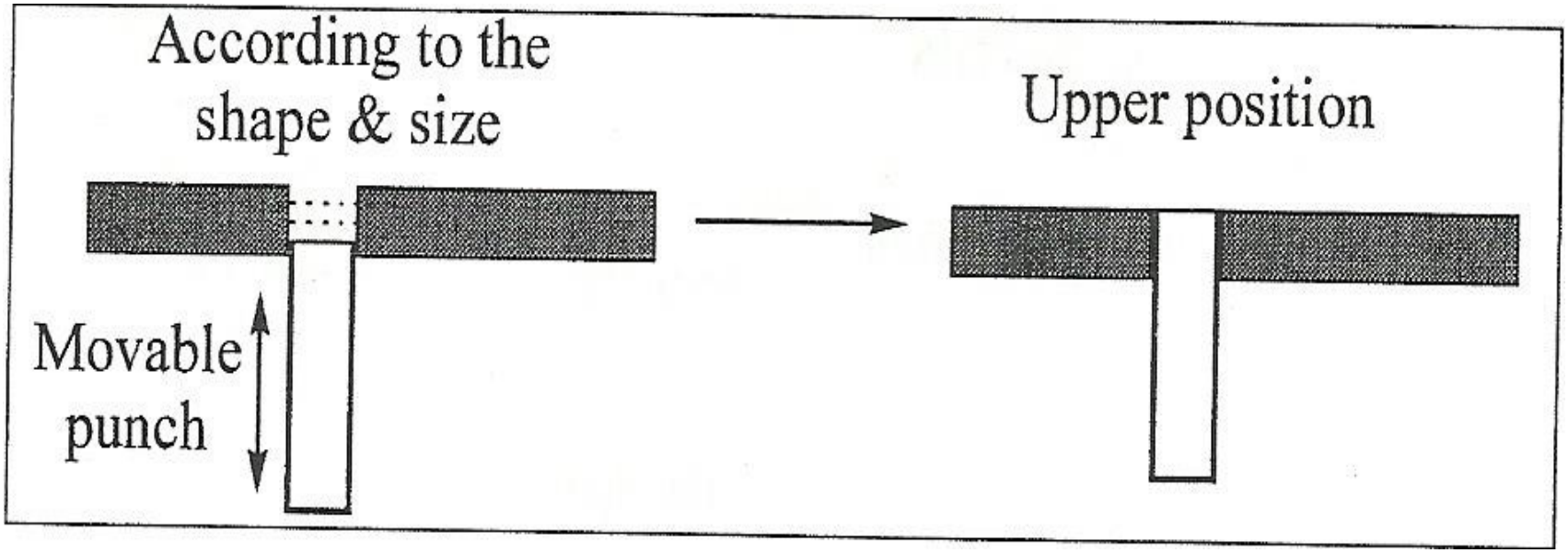
c) Two punches: for tablet pressing



N.B.:

- **The die cavity can be manipulated according to t shape and size.**

-The upper and lower punch heads have the same shape and dimension of the tablet, i.e. if the tabl is round⟶ the punch head is concave; if the tablet is flat⟶ the punch head will be flat.


According to the
shape & size
Upper position
Movable
punch

The cycle compression:

1 -Filling:

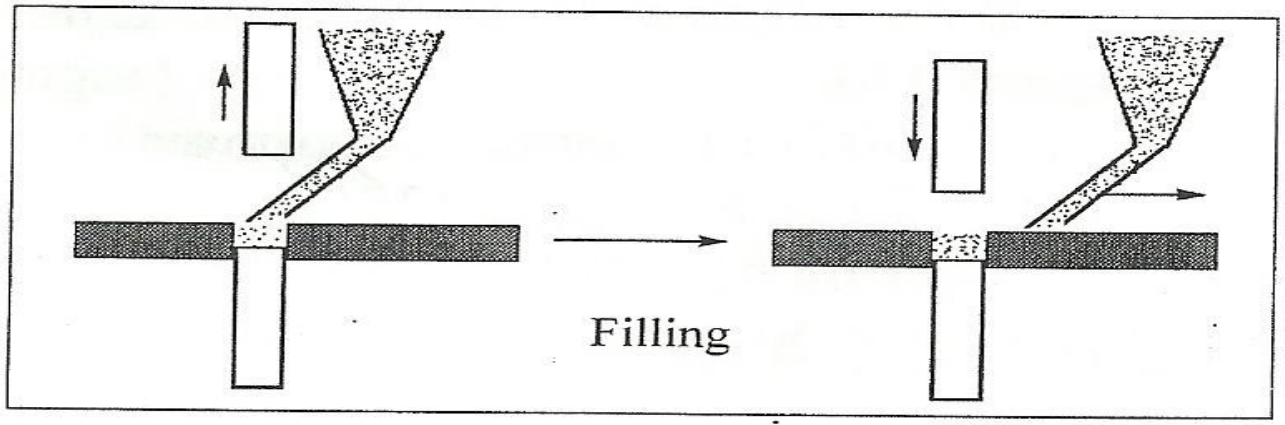
-Place the powder or granules in the hopper.

-The feed shoe moves towards the die to fill the

cavity with the powder.

-The feed shoe moves in back direction removing t

excess of the powder.

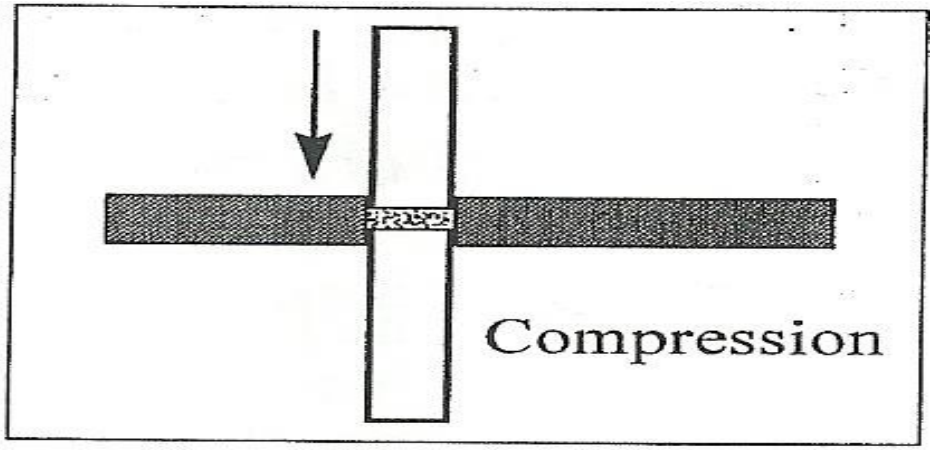


2 -Compression:

The upper punch descends into the die

compress the powder against the lowe

punch which remains stationary (fixed)

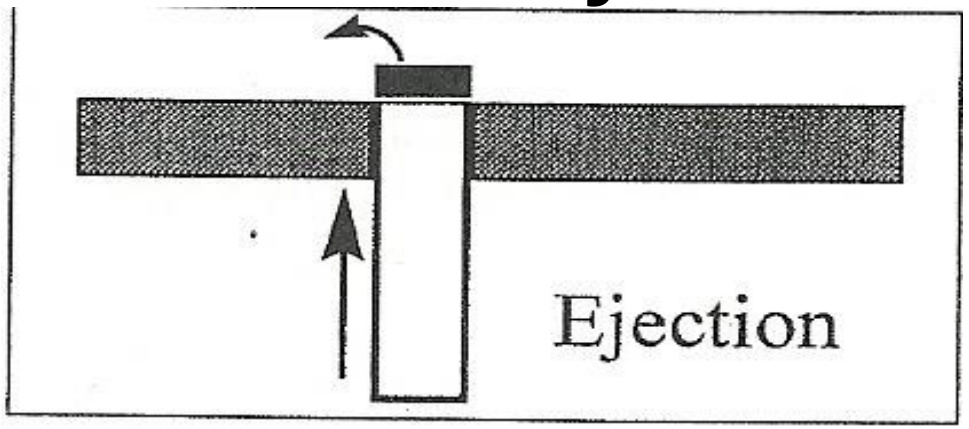


3 -Ejection:

The upper punch moves upwards, the lower p

rises and eject the tablet from the die cavity.

At this point, the feed shoe move in & knock the tabletout of the machine and the lower punch move down ready for the next cycle.



Formulation of tablets

A -Active ingredients (one or more ingredients)

Excipient(additives):

I -Excipient affecting the compression characteristic:

1 -Diluents

2 -Binders

3 -Anti-frictional agents: a) Lubricant

b) Glidants

II -Excipient affecting the biopharmaceutics of the drug and/or give additional physical properties of the tablet:

1 -Disintegrants
2 -Coloring agents

3 -Flavoring agents

4 -Sweetener

5 -Adsorbents.

I - 1 - Diluents:

They're inert substances which are added to drug in sufficient quantity to increase the bu of the tablet powder to a practical size for compression.

N.B.:It's difficult to compress powder less th 50mg, but if the drug dose is large as aspirin (300mg), there would be no need for a dilue

Types of diluents:

a) **Organic:**

1. **Lactose:It's water soluble, pleasant taste, neu reaction, non hygroscopic. It's available as cryst and as dried spray form (for direct compression).**
2. **Mannitol: It's expensive and when the table dissolves, a cool sensation in the mouth is pr Mannitolis mainly found in chewable tablets**
3. **Dextrose & sucrose: They're commonlyused as a combinationwith other diluents rather than alon because of their disadvantages(hygroscopic& no suitable fordiabetic persons).**

4. **Starch: It's used mainly as a disintegran glidant and not commonlyas a diluentin tablets because of its poor compressibil (mainly used as a diluentin capsules). To overcome this problem,modified com starc is used but it doesn't dissolve easily.**

5. **Microcrystalline cellulose: It's expensive, doesn't dissolve easily in water but it has a excellent compressibility and mainly used for direct compression.**

b)Inorganic:

1 -DicaIciumphosphate.

2 -Calcium sulphate.

They both have a good compressibility,

insoluble in water, non hygroscopic and not

expensive.

Selection of the diluents:
selection is based on:

1. 1. Experience of formulator & cost factor (mustn' expensive).
2. Compatibility with the drug and other excipients diluent must be inert otherwise interaction may For example:

-Ca salts + tetracycline ⟶ insoluble complex & poorly absorbed.

-Drug containing a primary amino group + diluent bearing a carbonyl group (lactose, dextrose & su ⟶ Colored product (Schiffsbase), so the drug will have on its surface brown patches.

3. **Initial moisture content of the diluent and its ab to absorb moisture.Moisture may have an adver effect on drug stability (hydrolysis) and physical properties of the tablet.**

Moisture can be:

-Unbound: aborbed from the environment

-Bound or water of crystallization: the condition under which water is liberated should be known (exsiccation & efflorescence). For example: dicalciumphosphate is non hygroscopic and has a bound moisture liberatedat 80 Candabove → Stable & recommended for moisture sensitive dr suchas vitamins.

4. **Solubility of the drug: when the drug substance has low water solubility so it's recommended to use a water soluble diluent to improve the dissolution rate of the drug.**

I - 2 - Binders:

They're adhesive materials used to hold the powder together as granules to have in the end a good flow & compressibility.

Types of binders:

a)Natural binders:

1. **Starch paste: or starch mucilage (5 –10% in water), it's the most general adhesive material (of choice) because it's inert & neutral.**
2. **Gelatin solution: (10 -20% in water) the gelatin must be freshly prepared and must used while it's warm to prevent the gelling process. It's a strong adhesive material so will give hard granules difficult to compres So gelatin is used as a binder in the case of tablets that don't need disintegrants as lozenges.**

3. **Gum mucilage: it must be freshly prepared It's mainly used in lozenges because, like gelatin, gum mucilage gives hard granules**

4. **Dextrose and sugar syrup: It's a good adhesive material but it's hygroscopic.**

b) Synthetic binder:

1. Polyvinylpyrrolidone or Povidone (PVP): it's one of the best general adhesive material (2nd choice). It's soluble in water and alc
2. Cellulose derivatives: they're good adhesiv material. Examples for cellulose derivative

-Methyl cellulose, hydroxypropylmethyl cellulose (HPMC), sodium carboxymethyl cellulose (Na CMC) ⟶ soluble in water and alcohol.

-Ethyl cellulose ⟶ insoluble in water but soluble in alcohol.

Method of addition:

1. Powder + moisturizing solvent then we ad
   dry binder → less effective.

2. Binder + moisturizing solvent then we add
   the powder → more effective Examples on
   moisturizing agents: water (most used), et
   & isopropyl alcohol and mix of water and
   alcohol.

1-3- Antifrictional agents:

I-3-a- Lubricants:

They're substances added to the granules to:

1. **Facilitate the ejection of the tablet from the die cavity because lubricants reduce the friction between the granules & die walls during compression and ejection.**

2. **Prevent adhesion of the tablet granules to the surface of punches.**

3. **May improve the flow properties of the drug.**

Types of lubricants:

a) Hydrophobic lubricants: (< 1 % in the formulation)

1. Metallic stearate(Mg, Ca, Zn)
2. Stearic acid
3. Waxes
4. Hydrogenated vegetable oil or Sterotex
5. Talc (5% in the formulation)

b) Hydrophilic lubricants: less commonly use

It's used only in water soluble tablets as effervescent tablets examples:

1. **PEG 4000 -6000**
2. **Magnesium laurylsulfate.**

Methods of addition:

Lubricants should be finely divided powder a

added to the powder by sieving then mixed g

for 2 to 3 min to distribute the lubricant to th

surface of the powder.

An excess amount or over mixing of lubricant
may adversely affect the dissolution rate plu
prolonged disintegration time may occur
because the lubricant is hydrophobic (low
solubility).

Selection of the lubricant:

The lubricant must be compatible with the drug,for example, for Aspirin if we use Mg stearate as a lubricant, it will enhance the decomposition of aspirin into salicylic acid (hydrolysis occurs due to the slight alkalinity the lubricant).

Solution: use aspirin with stearic acid or waxe as a lubricant.

I-3 -b -Glidants:

They're substances added to improve the flow properties of the granules. Its mode o

action is to reduce the inter-particulate friction by coming in between the surface the particles or by smoothing the surface

the particles.

Type of glidants:

1. **Colloidal siliconedioxide(0.2-5%) also called Cab-o-silor Aerosil.**

2.Starch5%

3.Talc 5%

N.B.:Colloidal silicone dioxide is the preferred one because of its effectiveness in low concentration.

Method of addition:

Added with the lubricant

II -1 -Disintegrants:

They're substances added to the tablet to facilitate the break up or disintegration af administration of the tablet. Disintegrants are used because the rate of dissolution o disintegrated particles is much greater th the intact tablet form.

Mechanism of action:

1-Swelling:

The disintegrants wells in contact with water

thus facilitating rupture of the tablet.

2 -Capillary action:

The spherical shape of the disintegrant increases the porosity of the tablet thus enhancing the penetration of water (or liquid
Capillary and swelling action may occur together.

3 -Producing gas:

This process occurs only in effervescent tablet and effervescent mix is citric acid + $NaHC0_3$.

Type of disintegrants:

1. **Starch:(10 -20%)⟶ Mode of action by swelling and capillary.**

   Potato and corn starch are widely used, they're good disintegrant and cheap.

N.B.:Starch should be present in 10 to 20% o the whole formulation; otherwise it's considered as a glidant and not as a disintegrant.

2.Super disintegrants:(2 -4%)

a) Sodium starch glycolateor Explotab: by swelling action.

It's a cross linked starch. It swells 10 times t

size of the tablet in few seconds.

b)CrossCarmelose: → by swelling action.

It's cross linked carboxymethylcellulose.

c) Cross Povidone: → by capillary action.

It's a cross linked povidone.

3. Cellulose derivatives:→ by capillary action

a) Microcrystalline cellulose or Avicel.

b) Carboxymethyl cellulose.

4.Clays:(10%)

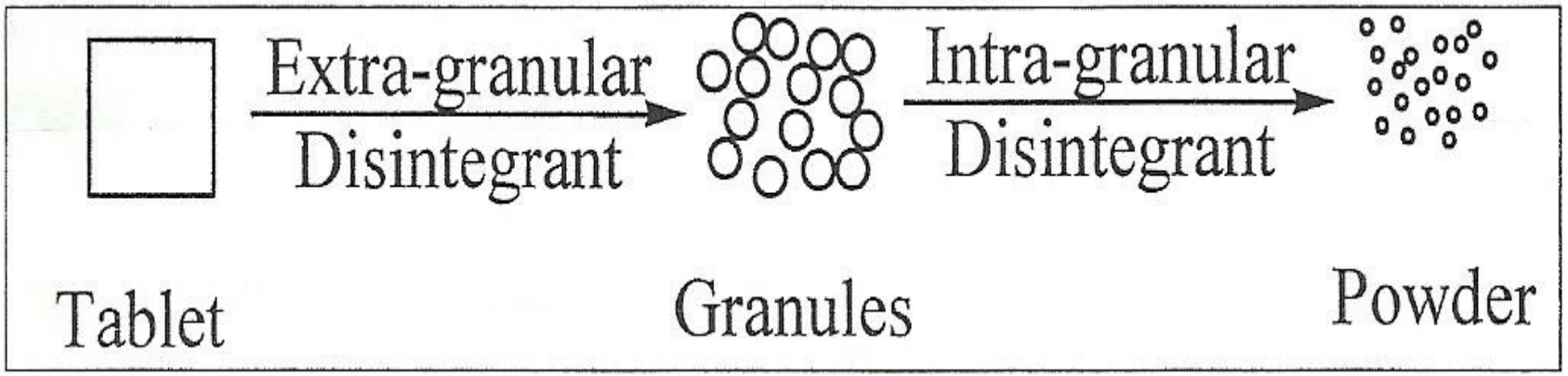
a) Bentonite.

b) Veegum.

c) Iolite or cordierite: It's a gray mineral mad
from silicate of magnesium, aluminum, an
iron. Its main disadvantage is that it loose
its disintegrant character upon wetting st
during granulation so it must be only used
a dry condition.

Method of addition:

1. **Intra-granular disintegrant:it's the part added before granulation.**
2. **Extra-granular disintegrant:it's the part added after granulation.**



II - 2 - Coloring agents:

They're substances added to the tablet to

1. **Mask the color of the drug.**
2. **Identify a product from another during manufacturing.**
3. **Identify a particular drug for a patient.**
4. **Produce a more elegant tablet.**

Types of coloring agents:

a. **Inorganic pigments:mineral pigments are used for solid dosage form and for externa used solutions as ferric oxide & Titanium dioxide.**

b. **Natural coloring agents:Caramel, Saffron (bright yellow), Chlorophyll (green), Xanthophylls (yellow), Riboflavin or VitB (yellow) and *β*-Carotene (orange).**

c. **Synthetic organic dyes:The synthetic color must be approved and certified by the FDA before usage. Plus they must not be used i pediatric preparations (only use natural colors). Coal tar dyes, Erythrosine (red), col lakes which are dyes adsorbed on Aluminu hydroxide layer (25 -35%); they're used to have a more uniform color distribution so they are more preferable than other dy**

Method of addition:two methods are used:

1st-Dispersed the color in granulating solution.

2nd-In direct compression: mix the lake with powder before compression.

II -3-Flavoring agents:

They're substances added to the tablet to mask its unpleasant taste.Flavoring agents are mainly used in effervescent and chewable tablets.

According to the taste:

a. **Salty taste:Liquorices (glycyrrhizin syrup) or Cinnamon.**
b. **Sour taste:Lemon flavor.**
c. **Bitter taste:Fruit syrup, Raspberry syrup, Chocolate syrup.**
d. **Natural flavoring agents:aromatic oil, spirits, liquid extract, tinctures, aromatic water.**

e. Synthetic flavoring agent: *Methyl salicylate* (oil of winter green), *Benzaldehyde*(bitter almond), *Vanillin*.

According to the nature:

a. Phenols(oilorspray dried form): Clove, vanillin, methyl salycilate.

b. Aldehydes:Cinnamic acid or oil, citral of lemon, benzaldehyde.

Method of addition:

1 -If oil is used: dissolve the oil in a volatile solvent & sprayon granules.

2 -If spray is used: mixing or spreading with the powder or granules just before lubrication step.

II -4 -Sweetening agents:

They're substances added to effervescent chewable tablets when the commonlyused diluents such as sugar, mannitol, lactose & dextroseare not present. Examples: natural sugars & artificial sugars such as Aspartame

(180 times sweeter than ordinary sugar), saccharin(500timessweeter)and cyclamates.

II -5 -Adsorbents:

They're substances which can carry a large amount of liquid without swelling, Example silicone dioxide can carry an amount of liqu equal to 50% of its own weight.

Adsorbents are added when the tablet formulation contains:

-Oil (in multivitamins).

-Liquid extracts.

-Hygroscopic ingredients.

Types of Adsorbents:

1. Silicone dioxide
2. Mg silicate
3. Mg oxide
4. Clays such as Kaolin.

Method of addition:

The hygroscopic ingredients are first mixed with the adsorbent and then added to the powder mixture.

Preparation of tablets

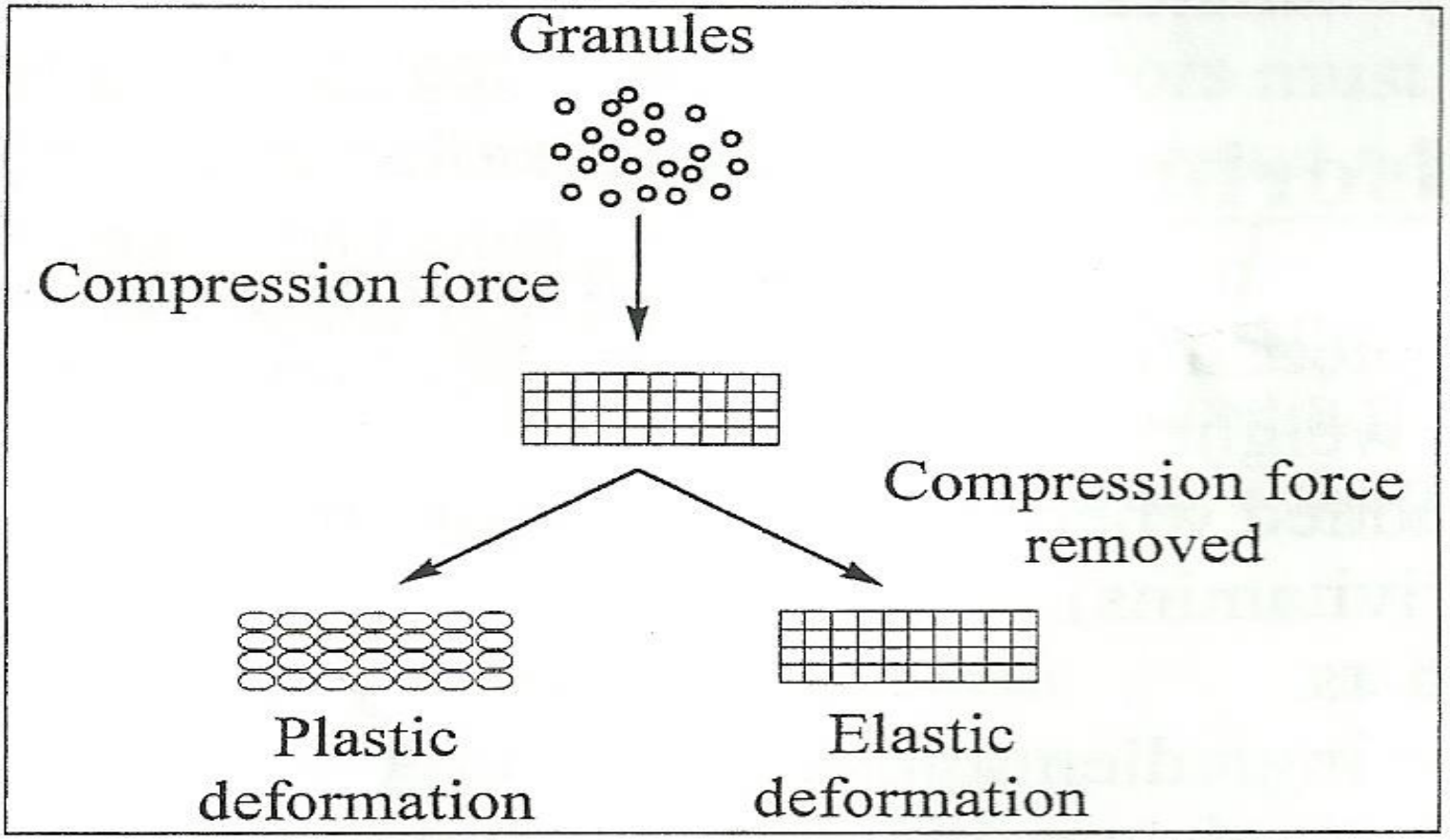
Before formation of perfectly compressed tablets, the powder must posses the followin properties:

-Good flow → Uniform filling of the die cavity resulting in uniform tablet weight.

Good compressibility → upon compression deformation of the particles will occur. After removal of the stress, the particles will exhib the following deformations:

a) Elastic deformation: cohesion force is lost and the particles try to retain its original shape and form.

b) Plastic deformation: the particles stay deformed & cohesion is retained.


Granules
Compression force
Compression force removed
Plastic deformation
Elastic deformation

Most of the particles exhibit both deformatio forms but one of them predominate the othe

If plastic deformation predominates → direct compression is used.

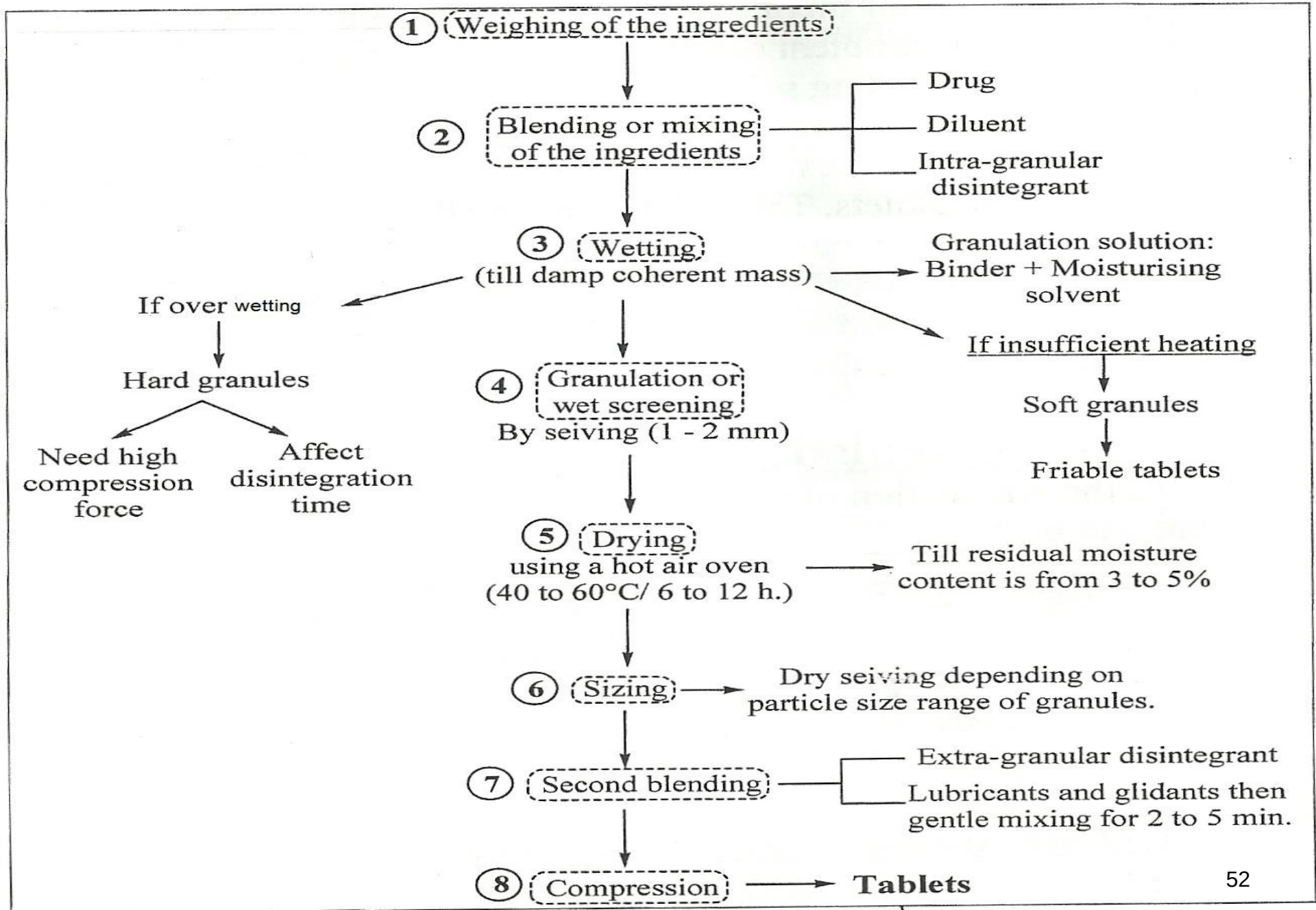
If elastic deformation predominates → plastic binders are used (granulation) or plastic diluents as Avicel are used.

According to the last information, the metho
preparation of tablets are:

I -Granulation:
a -Wet granulation
b -Dry granulation (least commonly used)

II -Direct compression

I – a – Wet Granulation:



Sweeteners and adsorbents may be added in 2nd step (first blend). Coloring agents may be added in the 3rd step (wetting).

Flavoring agents may be added in the 7th step (second blend) after the disintegrant.

The Advantages of wet granulation:

1. Changes the fine powder to free flowing granules with good compressibility & binding properties.
2. Suitable for large dose drugs having poor flow &compressibility.
3. Good distribution & uniform content for soluble small dose drugs; plus good color distribution is obtained if dissolved in the binding solution.
4. The dissolution rate of the hydrophobic drugs may be improved by wet granulation with the proper choice of binders and diluents.

The Disadvantages of wet granulation:

1. It's an expensive and long process.
2. It's not suitable for thermo-labile drugs as Aspiri (the problem of moisture may be overcome by using alcohol in the binding solution)
3. During the drying process, water soluble dyes ca migrate to the surface of the granules causing unequal distribution of color on the surface of th tablets. This effect is called the Mottling effect. Mottling effect can be overcome by:

   a) Using insoluble lakes.

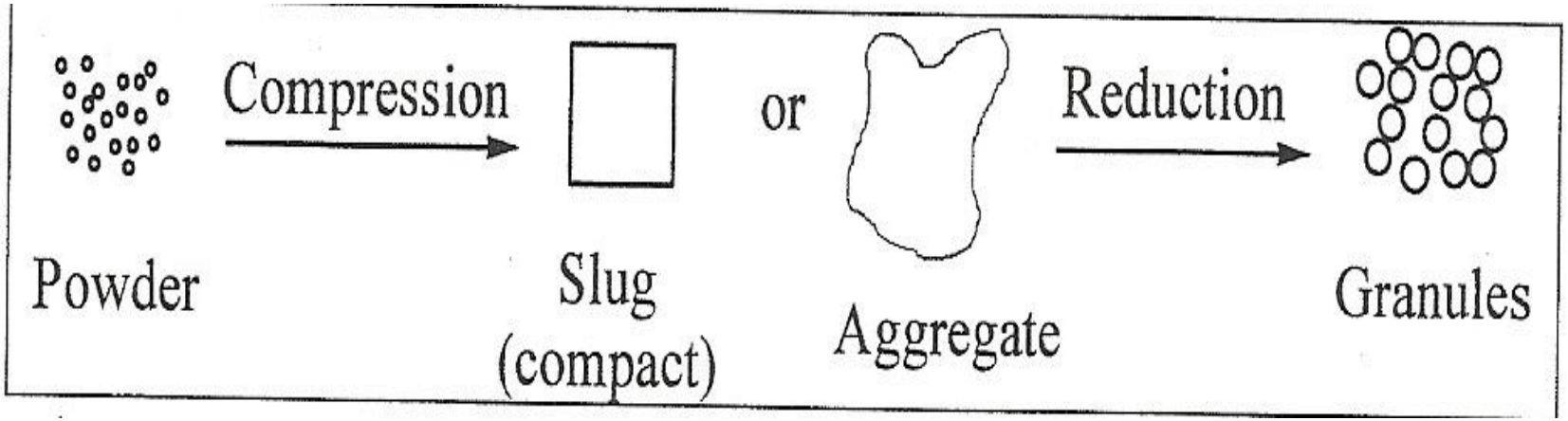
   b) Increasing the viscosity of the binding solution.

   c) Adjusting the drying rate.

I - b - Dry granulation:

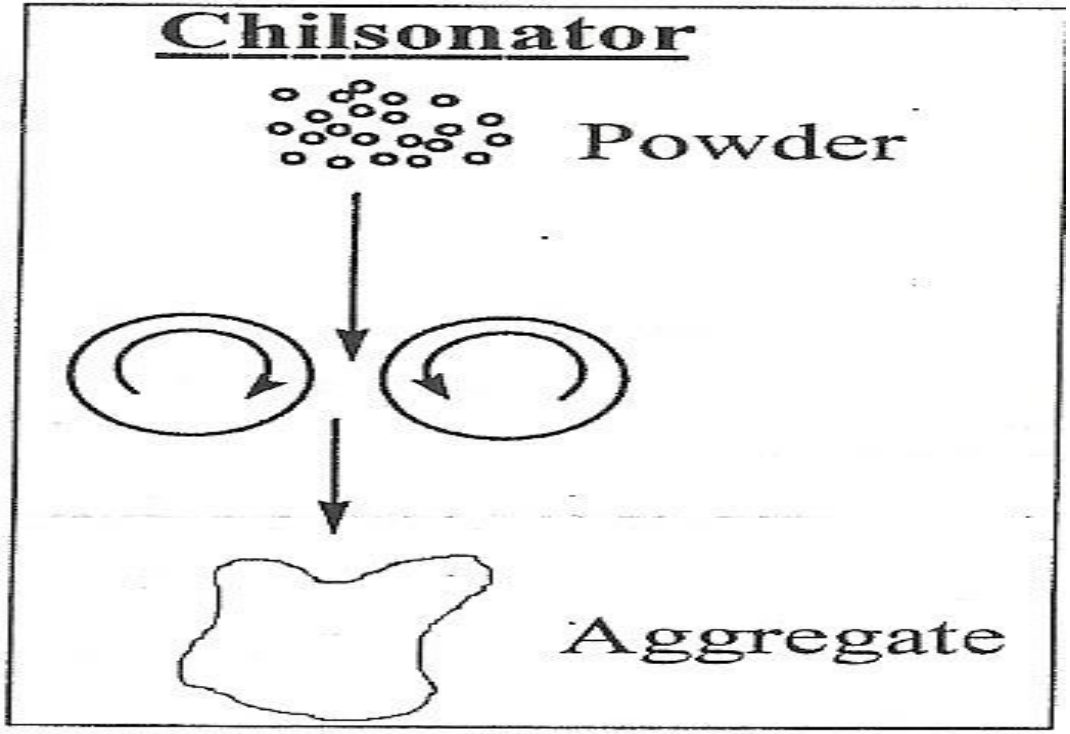
It's the granulation of the powder by compression without theuse of solvent and heat.

Note that the powder should have certain cohesive properties to prepare the compact form.



Method of preparation of the compact form:

1. **Slugging:using an ordinary tabletmachine It consists of a large flat faced punches yielding to a 2 or3 cm tablet in diameter with a great thickness.**
2. **Pressure roll process:gives large aggregat**

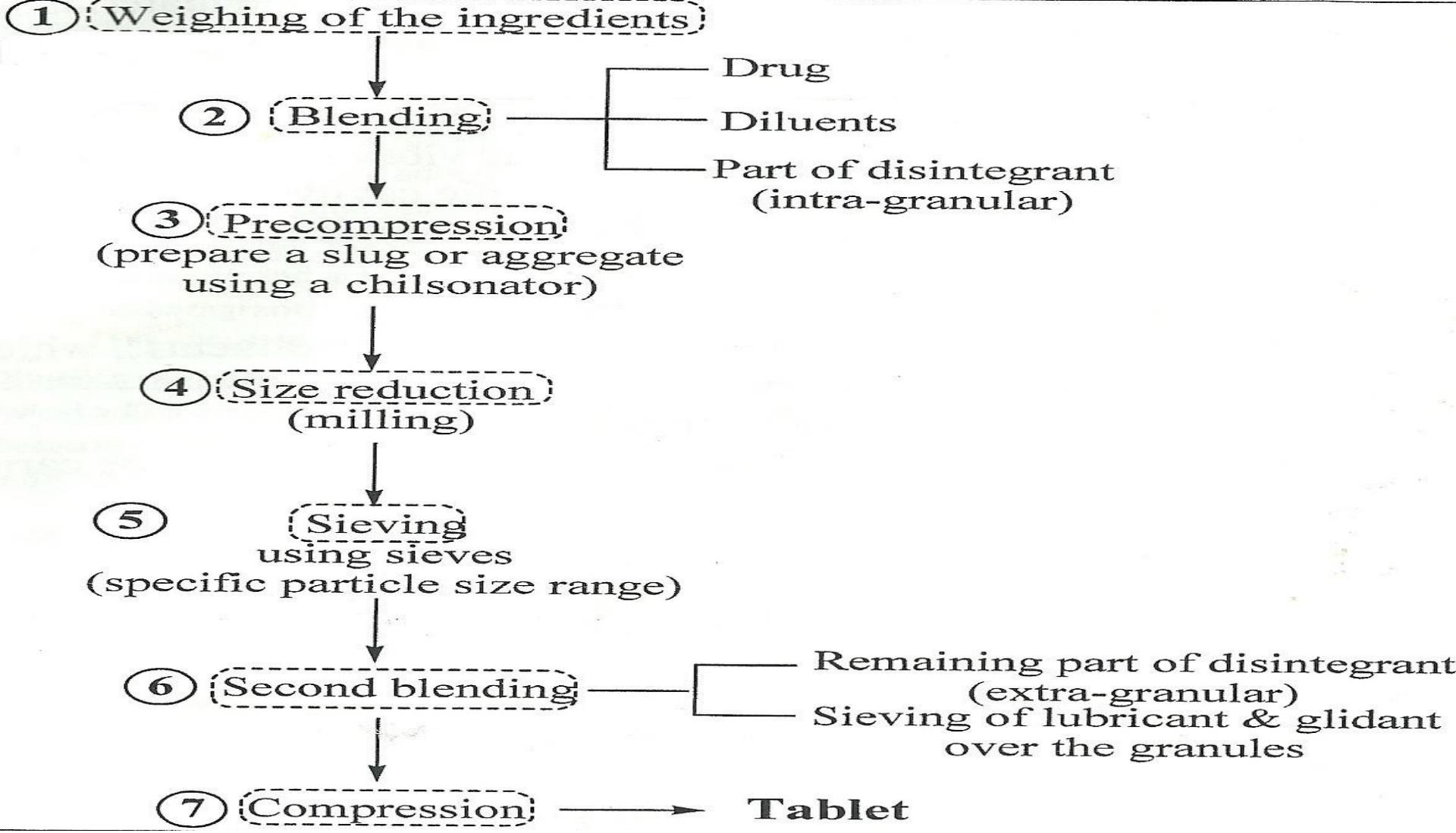


The Advantages of dry granulation:

Suitable for heat& moisture sensitive drugs as in effervescent tablets (heat sensitive) and aspirin tablets(heat & moisturesensitive)

The Disadvantages of dry granulation:

1. This method is effective only with a limited number of materials which posses some cohesive prop
2. It requires specialized heavy machine such as Chilsona
3. Does not permit uniform color distribution.



Sweetening and coloring agents may be added in th

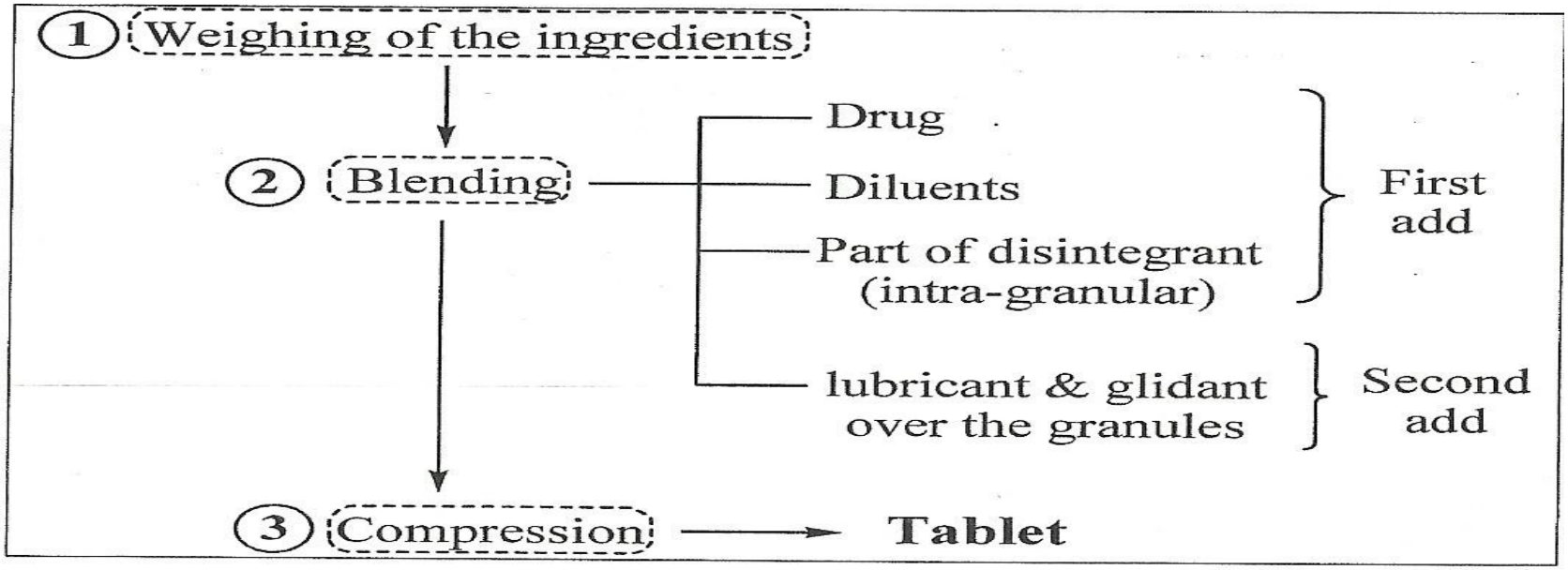
2nd step (first blend), Flavoring agents may be added

the 6th step (second blend),

II -Direct compression: The powder mixture i

directly compressed into the tablets without

pretreatment.



This method is reserved for limited number c crystalline material such as Na or K halides because most of the materials have inherent *poor* compressibility and flowability.

A -To improve the flow properties of the materials:

1. **Usage of force feed devices (as Ultrasonic vibrator) which repel the air from the pow and force it through the die cavity.**
2. **Add glidants(with or without the force fee device).**

B -To improve the compressibility of the materials:

Use direct compressible fillers (plastic fille diluents), which have the same propertiesas ordinary diluents used, plus:

a -Good compressibility and flow properties

b -High dilution capacity: the amount of the drug which can be carried by the diluents i direct compression.

These Properties are imputed to the commor diluents by special processing steps such as granulation, crystallization & spray drying.

Examples of direct compressible fillers:

1. **Spray dried lactose → Processed by spray drying. Disadvantage: Low dilution capacit**
2. **Granular mannitol → Processed by granulation.**
3. **Compressible sugar → Processed by crystallization.**

4. **Modified or compressible com starch (Sta-Rx 1500starch). It's made by partial hydrolysis of starch. It's composed of intact grain of starch spherical aggregate of hydrolyzed grain (brin out amylose & amylopectin which have good adhesive properties). Microcrystalline cellulo**
5. **Avicel: non fibrous**

   form of cellulose→ Processed by spray drying It's the best compressible filler because it has high dilution capacity and acts as a disintegra at the same time (so no need to add a disintegrant)
6. **Unmilleddicalciumphosphate or Em-compressed→ Processed by dry granulation.**

Diluentcapacity:
Avicel> Em-compressed> Modified starch>

Granular mannitol> compressed sugar > Spra

dried lactose.

Because Avice! is very expensive, a mix of Av

with other diluents as modified com starch o

spray dried lactose is used.

Formulation of tablets made by direct compression

The formulation of tablets made by direct compression depends on the drug quantity:

A. Tablets containing small drug quantity: (<25% of total tablet mass).

1. **Direct compressible fillers (no problem we used spray dried lactose)**
2. **Glidants**

B. Tablets containing large amount of drug:

1. Direct compressible filler (high dilution capacity filler such as Avicel).
2. Force feed devices.
3. Choose the particle size range and crystall form of the drug which have some adhesiv properties.

The disadvantages of direct compression:

1. **Drugs characterized by high dose & high bulk volume having poor compressibility and fluidity can't be prepared by direct compression. Instead we use wet granulat method.**

2. **The cost of raw materials and its testing process are high: the physical properties o raw material such as particle size, fluidity, moisture content and crystal form are mo critical in direct compression than in wet granulation.**

3. It doesn't permit a uniform color distribution.

4. Direct compression powder mixture (drug + excipient) are subjected to seggregation yielding non uniform tablet content becau of the:

   a) Difference in particle size & density between the drug & excipient.

   b) Lack of moisture in the powder may giv rise to static charges which can lead to demixing.

To overcome these problems, try to keep the powder in the uniform particle size and dens

Properties of the core:

1. **Must have a convex surface.**
2. **Hard enough to withstand the coating process.**
3. **Disintegrate rapidly when coat dissolves.**

Types of compressed tablets

According to the route of administration, the compressed tablets are classified into:

I. **Tablets administered orally:**

1-Conventional tablets

2-Coated tablets

3-Multiple compressed tablets:

a) Multi-layer tablets

b) Compression coated tablets

4-Chewable tablets

II-Tablets used in the oral cavity:

1- Sublingual tablets

2-Buccal tablets

3-Lozenges

III-Tablets used to prepare solution:

1-Effervescent tablets

2-Dispensing tablets

IV-Tablets used by other routes:

1-Vaginal tablets

2-Implantation tablets (sterile tablets)

Tablets administered orally:

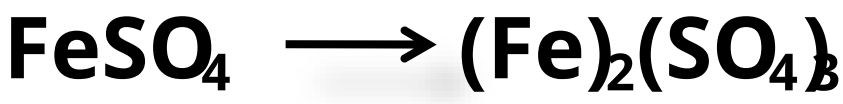
I-2-Coated tablets:

The general reasons for coating process are:

1-to mask the unpleasant taste, color and od of the drug.

2-to protect the drug from environmental conditions such as:

a) Air(auto-oxidation). Example:



b) Moisture

c) Light

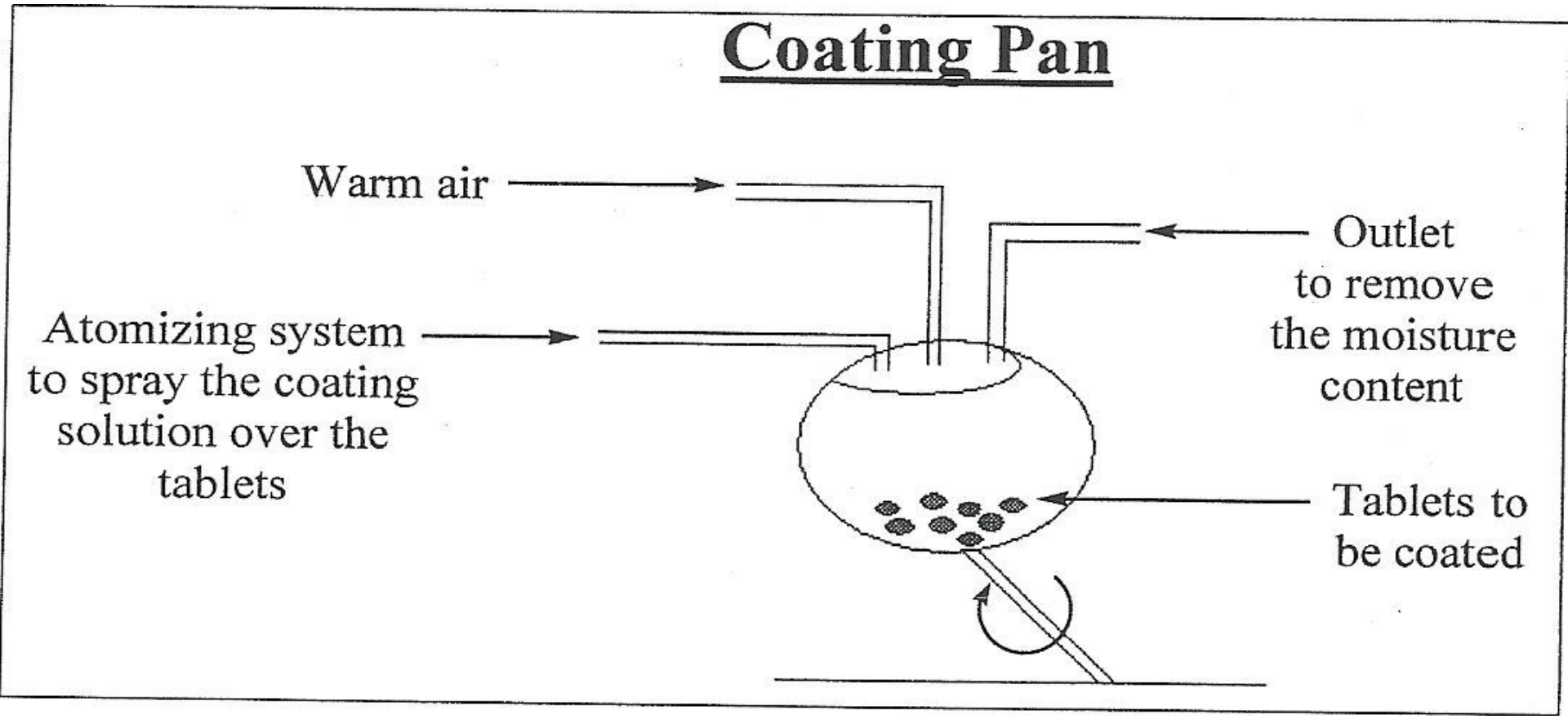
3-To control the release of the drug from the tablet.

4-To separate two incompatible drugs.

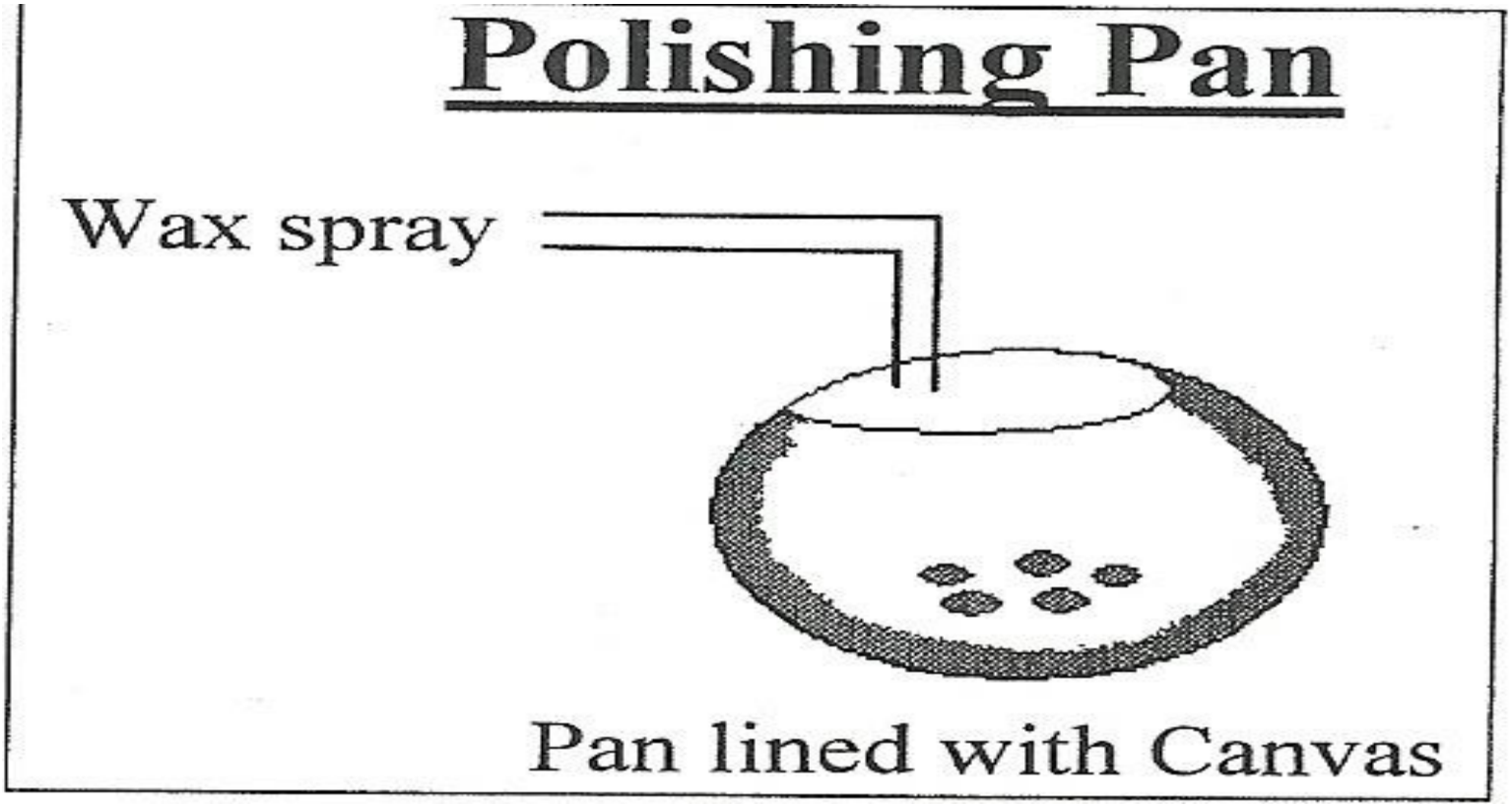
5-To protect the drug from gastric fluids.

Coating equipment:

1 - Coating Pan:



2 - Polishing Pan:



Types of coating:

1. **Sugar coating**
2. **Film coating**
   Compression coating.
3.

1- Sugar coating:

It's the process which involves building up layers of coating material on the tablet core they are rotated in a revolving pan. This is do after several application of the coating dispersion & drying of the solvent.

Sugar coating involves 5 steps:

1- Sealing step:

We use a polymer solution to form a film around the core to prevent the water from penetrating tablet during sugar coating.

Sealing polymer solution used are:

A.Cellulose acetate phthalate or CAP

B.Polyvinyl acetate phthalate or PVAP

C.Zeinused sometimes as binder

D.Shellac (resinous material) + PVP

Method of addition:

Spraying of the alcoholic sealing polymer solution is done in the rotating pan, then drying process takes place.

2- Sub-coating step:

It's used to round the tablet edges using sub-coating syrup made of:

a) Simple syrup with gelatin

b) Sub-coating powder: Talc, TiO_2 , starch, Calcium phosphate, $CaCO_3$, kaolin

Method of addition:

—Add the sub-coating syrup to the tablets until they become sticky

—Dusting the sub-coating powder over the stick tablets

—allow the pan to rotate until the tablets are dr

N.B.:Repeat for 3 to 4 times until the edges of th tablets are covered.

3- Smoothing step:

this step is done using simple syrup contain

1-5% TiO_2 as an opicifier(opaque).

We render the surface of the coat smooth

which is done by applying several layers of t

syrup & then drying. We repeat this cycle ur

we obtain a coat of a desired thickness and

smoothness.

4- Coloring step:

This step is done using colored syrup:

Simple syrup + dyes (up to 50 applications)

+ lakes (best: few applications)

+ inorganic pigments (few applications needed) Several applications of colored syrup until the required color is obtained, drying process is done after each application.

N.B.: if we don't need any colors, we repeat t smoothing step for several times becaus

Ti0₂is whiteincolor.

5- Polishing step:

The coated tablets are polished by rotating t

in a polishing pan which contains a special

tissue (Canvas) with the addition of a volatile

solution of wax as Bees wax.

2 - Film coating:

Film coating involves the deposition of a thin flexible uniform film of polymer on the surface of the tablet. The procedure involves only one step: spraying the film coating solution over tablets and then drying them using a stream air.

The film coating solution contains:

1. **Film coating polymer**
2. **Plasticizer**
3. **Coloring agents**
4. **Flavoring & sweetening agents**
5. **Solvent**

1- Film coating polymer:

a) Properties:

-The polymer should be soluble in various solvents.

-The film should be soluble in the gastro-intestinal fluids.

-The film should produce a strong continuou flexible and elegant film (the most import character).

-The film should be stable in different environmental conditions (air, moisture, heat, light. ..) and should be compatible w

b) Classification:

According to the solubility in the gastro-intestinal fluid we identify:

1 -Non-enteric film coating polymers:

this type of polymer dissolves in the G.I.T. fluids (dissolves in acidic & basic medium).

Example: cellulose derivatives: HPMC, HPC, N carboxymethylcellulose and methyl cellulose

2 -Modified release polymers:

i) Enteric coating polymers:

It's insoluble in the gastric fluids & remains intact in the stomach but it dissolves & relea its content once it reaches the small intestin

This type of polymer is used to:

» Protect the gastric mucosa from the side effec

the drugs as NSAID (Ketoprofen, Indomethaci

» Protect the drugs which are unstable in acidic

as Erythromycin.

» Prevent the local defect in the stomach and to

deliver the drug into the intestine (site of acti

in a concentrated form to have a systemic or l

effect (anthelminticor intestinal antiseptic).

Examples of enteric coating polymers:

1. Cellulose acetate phthalate or CAP:

Its main disadvantage is that it dissolves at p

= 6 which delays the release of the drug long

than it is desired because most of the drugs a

absorbed in the first part of the intestine at

pH = 5.

Plus it is heat & moisture sensitive and it's

decomposed into acetic and phthalicacid.

2. Hydroxy-propylmethylcellulose phthalate

HPMCP:

It's a stable polymer which dissolves at pH =

and it is the most used polymer.

3. Polyvinyl acetate phthalate or PV AP:

It's a stable polymer which dissolve at pH = !

ii) Sustained or controlled release film polymers:

There are two types:

1. **Slowly dissolves in the gastro-intestinal fluids as a mix of waxes with stearic acid or with glycerylmono stearate**
2. **Insoluble in the gastro-intestinal fluids as e cellulose. It works by capillary action by permitting the water to penetrate inside th film membrane which will dissolve slowly t drug and then diffuse it out again.**

iii) Enteric & sustained polymers:

They dissolve at pH = 6 -7 and posses both

enteric and sustained properties of a coatin

polymer. Example: Acrylate polymer.

2.Plasticizer:

They're used to:

A.Increase the flexibility of the film

B. Prevent the adhering of the tablet to the walls of the coating pan.

C. Prevent the tablet from sticking to each o

There are two types of plasticizer:

i) **Aqueousplasticizer:**

-PEG 200 -400

-Glycerin

-Propylene glycol

ii) Non-aqueous plasticizer:

-Castor oil

-Di-butyl sub-acetate

Method of addition:

We add the plasticizer with the polymer in a 1:10 ratio.

3. Coloring agents: same coloring agents use
before.
4. Flavoring & sweetening agents: same ager

used before.
5. Solvent:

a) Water

b) Organic solvents: Ethanol, isopropanol,
acetone, methylenechloride

Comparison between sugar and film coating:

	Sugar Coat	Film Coat
1	Made from syrup	Made from polymer
2	Includes 5 steps	Includes only one step
3	Operator training is required	Operator training is not required
4	The coat will increase the weight of the tablet by 50% High degree of polish	The coat will increase the weight of the tablet by 2 to 3% Not as shiny as sugar coat.
5 6	Its function is to protect the drug from environmental conditions	Its function is to protect the drug from diverse conditions (protection of gastric mucosa ... plus environmental conditions On aging no adverse effect occurs w... the disintegration time.
7	On aging prolongation of disintegration time occurs because the syrup contains acacia & gelatin which harden on time.	

1-3 multiple compressed tablets

1- 3 - a - Multi-layer tablets:

They're composed of 2 or 3 layers of granulations compressed together one on th top of the other. These layers have different colors.

Usage of multi layer tablets:

1. **To separate two incompatible drugs by an inert layer:**